2 **(a)** the energy per unit mass required to change a substance from the liquid phase to the gas phase without a change of temperature

(b) **1.** increasing separation of molecules / breaking or overcoming (intermolecular) forces between molecules / overcome potential barrier

2. doing work against atmosphere (during expansion)

(c) **(i)** Assuming no heat lost to surroundings,

Heat supplied by water = heat gained by ice

$$m_{\text{water}}\ c\ \Delta\theta = m_{\text{ice}}\ l_f + m_{\text{ice}}\ c\ \Delta\theta$$

$$900 \times 4200 \times (85 - \theta) = 100 \times 3.34 \times 10^5 + 100 \times 4200 \times \theta$$

$$\theta = 68.5°\text{C}$$

(ii) For (the same mass of) steam and water to reach thermal equilibrium with the skin, there is an extra amount of thermal energy released during condensation (from steam to water).

(d) **(i)** Area under graph (against the y-axis) = PΔV

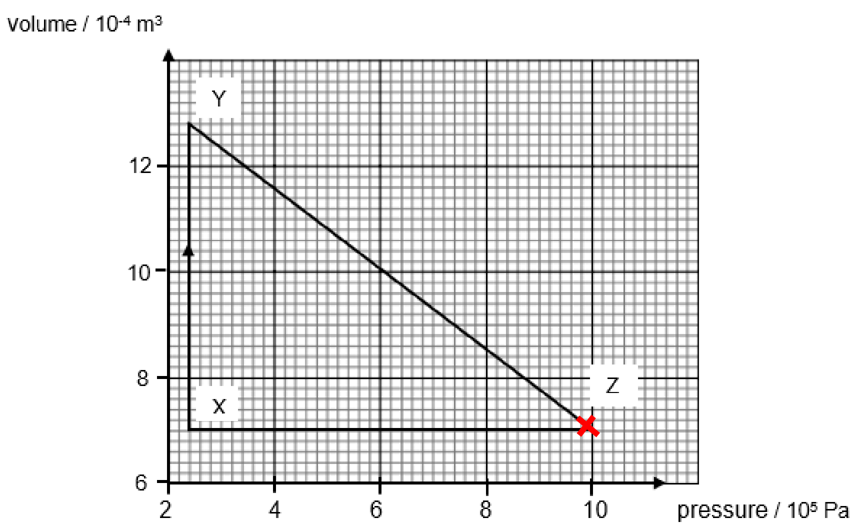
$= (2.4 \times 10^5)(12.8 - 7.0) \times 10^{-4}$

$= 139$ J

Work done on gas = − 139 J

(ii) No work done ZX, by First Law of Thermodynamics / $\Delta U = Q + W$, since if there is decrease in internal energy, there must be heat removed from the system.

(iii)



At $(9.8 \times 10^5, 7.2 \times 10^{-4})$.

At that point (marked x), the PV product is the largest.

3(a)	Separation: No change, since λ, D and a are constants